3. **Pollination by Animals**

When pollination takes place with the help of animals, it is called **Zoophily**. Flowers of such plants attract animals by their bright color, size, scent etc. e.g. sun bird pollinates flowers of *Canna, Gladioli* etc., Squirrels pollinate flowers of silk cotton tree.

## 17.6 Fertilization in Plants

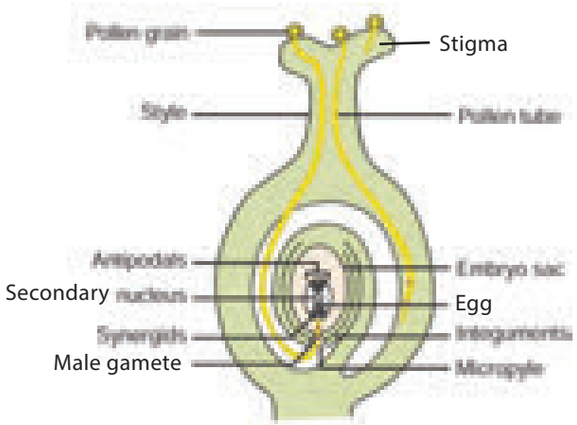



**Figure 17.12** Process of Fertilization

- Pollen grains reach the right stigma and begin to germinate.
- Pollen grain forms a small tube-like structure called pollen tube which emerges through the germ pore. The contents of the pollen grain move into the tube.
- Pollen tube grows through the tissues of the stigma and style and finally reaches the ovule through the micropyle.
- Vegetative cell degenerates and the generative cell divides to form two sperms (or male gametes).
- Tip of pollen tube bursts and the two sperms enter the embryo sac.
- One sperm fuses with the egg (syngamy) and forms a diploid zygote. The other sperm fuses with the secondary nucleus (Triple fusion) to form the primary endosperm nucleus which is triploid in nature. Since two types of fusion syngamy and triple fusion take place in an embryo sac the process is termed as **double fertilization**.
- After triple fusion, primary endosperm nucleus develops into an endosperm.
- Endosperm provides food to the developing embryo.
- Later the synergids and antipodal cells degenerate.

### Significance of Fertilization

(i) It stimulates the ovary to develop into fruit.

(ii) It helps in development of new characters from two different individuals.

**Post fertilization changes:**

1. The ovule develops into a seed.
2. The integuments of the ovule develop into the seed coat.
3. The ovary enlarges and develops into a fruit.

The seed contains the future plant or embryo which develops into a seedling under appropriate conditions.

## 17.7 Sexual Reproduction in Human

You have studied the structural details of the male and female reproductive system in 9th standard. In human beings the male and female reproductive organs differ anatomically and physiologically. New individuals develop by the fusion of gametes. Sexual reproduction involves the fusion of two haploid gametes (male and the female gametes) to form a diploid individual (zygote).

Organs of the reproductive system are divided into primary and secondary (accessory) sex organs.



10th_Science Unit-17.indd 248 11-01-2020 15:15:40